HOW TO USE THIS TEMPLATE (delete this box before submitting)

- Keep every heading and its order. Replace the grey italic guidance and every [bracketed placeholder] with your own text.
- Add or delete the member rows in the tables (title page, declaration, Appendix A) to match the size of your group.
- Format: A4, Times New Roman 12 pt body, 1.5 line spacing, 1-inch margins; headings as already set in this file.
- Length: 12-20 pages from Chapter 1 to the References (title page, declaration, contents and appendices not counted).
- Number and caption every figure ("Figure 3: ...", below it) and table ("Table 2: ...", above it), and refer to each in the text.
- Cite sources in IEEE style: [1], [2] in the text, full entries in the References.
- Before exporting to PDF, right-click the table of contents and choose Update Field > Update entire table.

LOVELY PROFESSIONAL UNIVERSITY

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CA-1 Project Report

[Project Title - exactly as in the CA project list]

Course: [Course Code] - [Course Title]

Section: [Section] Group: [G No]

Student Name	Registration No.	Roll No.
[Member 1 name]	[Reg. No.]	[Roll No.]
[Member 2 name]	[Reg. No.]	[Roll No.]

Problem source: [Competition / hackathon name], [PS ID if any]

Submitted to: [Instructor name]

Date of submission: [DD-MM-YYYY]

Declaration

We declare that this report and the accompanying implementation are our own work, carried out for the CA-1 project of [Course Code]. All data, code, papers and websites we used are acknowledged in the References. All results reported here were produced by our own implementation.

Use of AI tools: [State "None", or name the tool(s) and what they were used for, e.g. "ChatGPT to debug a plotting error in eda.ipynb; all analysis and writing are our own."]

Name	**Registration No.**	**Signature**
[Member 1]	[Reg. No.]	
[Member 2]	[Reg. No.]	

Abstract

150-250 words, one paragraph, written last. State the problem, the data, the method, the baseline, the main result with its number (e.g. "F1 of 0.84 against 0.71 for the baseline"), and what the system delivers.

Keywords: [3-5 keywords]

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1. Introduction

1.1 Background and Motivation

Explain the real-world setting in plain language and why the problem matters. Half a page.

1.2 Problem Statement

Copy the problem statement from your CA project allocation, then explain it in your own words in 3-5 sentences.

1.3 Source of the Problem

Name the competition or hackathon, the organiser, the problem-statement ID (for SIH) and the official link, exactly as in the allocation. Say clearly that your project is inspired by this problem - not a full solution to the competition.

1.4 Who Benefits

Identify the users or stakeholders and how your system would help them.

2. Objectives

Copy your three objectives from the CA project allocation, unchanged and numbered 1-3. Under each, add one or two lines on how you will show it has been met (which experiment, metric or demo).

3. Related Work

Briefly review 3-5 existing approaches: research papers, competition write-ups or existing tools. For each, say what it does and how your project differs or builds on it. Cite every source.

4. Dataset

4.1 Source and Licence

Name of the dataset, link, who published it, and its licence or terms of use.

4.2 Description

Number of records, features or columns, classes or targets, time span, and any class imbalance. Use a table.

4.3 Exploratory Data Analysis

Two to four figures (distributions, correlations, sample images or texts) with a sentence on what each shows.

4.4 Preprocessing

Cleaning, handling of missing values and outliers, encoding, scaling, and the train / validation / test split (with sizes).

5. Methodology

5.1 Overall Approach

A block diagram of the pipeline from raw data to output, with a short explanation of each block.

5.2 Baseline

The simple method you compare against (e.g. majority class, linear model, rule-based method) and why it is a fair baseline. This is mandatory.

5.3 Proposed Method

The model(s) or technique(s) you use, why you chose them, and the key settings or hyperparameters and how you selected them.

5.4 Syllabus Mapping

A table linking each technique you used to the course unit it comes from.

Table [n]: Mapping of project techniques to the course syllabus

Technique used in the project	Course unit	Where used (section / file)
[e.g. Standardisation]	[Unit 2]	[4.4 / src/preprocess.py]
[...]	[...]	[...]

6. Implementation

6.1 Tools and Libraries

Language, libraries with versions, and hardware (laptop / Colab / GPU).

6.2 System Architecture

How the code is organised (refer to the folder structure in Appendix B) and how the parts connect.

6.3 Demo / User Interface

Screenshots of the working system (e.g. Streamlit app) with captions, and what a user can do with it.

6.4 How to Run

Two or three lines pointing to the README in your code folder; full steps belong in the README.

7. Results and Evaluation

7.1 Evaluation Setup

Which data the results are on (test set / cross-validation) and the metrics, as named in your objectives.

7.2 Results Against the Baseline

A table with the baseline and your model(s) on every metric. Put the best value in bold. Use only numbers produced by your own code.

Table [n]: Results on the test set

Model	[Metric 1]	[Metric 2]	[Metric 3]
Baseline: [name]	[value]	[value]	[value]
Proposed: [name]	[value]	[value]	[value]

7.3 Error Analysis

Where does the model fail? Show a confusion matrix or examples of wrong predictions and explain likely causes.

7.4 Discussion

Interpret the results: did you meet each objective? What worked, what did not, and why?

8. Limitations and Ethical Considerations

Limitations of the data, method and evaluation. Any risks of bias, privacy or misuse, and how they could be reduced.

9. Conclusion and Future Work

Summarise what you built and achieved in one paragraph. Then list next steps, including the advanced extension from your allocation if you did not complete it.

References

IEEE style, numbered in order of first citation. Include the competition / problem-statement page, the dataset, papers, and any code you adapted. Example: [1] A. Author, "Title of paper," Journal, vol. 1, no. 2, pp. 1-10, 2024.

Appendix A: Individual Contribution

Every member must contribute to the code and to the report. Be specific; this is checked during the presentation.

Member	Main contributions (code, experiments, report sections, slides)	Share (%)
[Member 1]	[e.g. data cleaning, baseline, Sections 4 and 5]	[50]
[Member 2]	[e.g. model training, demo app, Sections 6 and 7]	[50]

Appendix B: Code Repository and Folder Structure

GitHub link (or "submitted as ZIP to the instructor") and the folder tree of your project, following the structure in the submission guidelines.

Appendix C: Additional Results

Optional: extra figures, longer tables, additional screenshots. Keep the main argument in Chapters 1-9.